Kalshi Prediction Market Inefficiency Investigation

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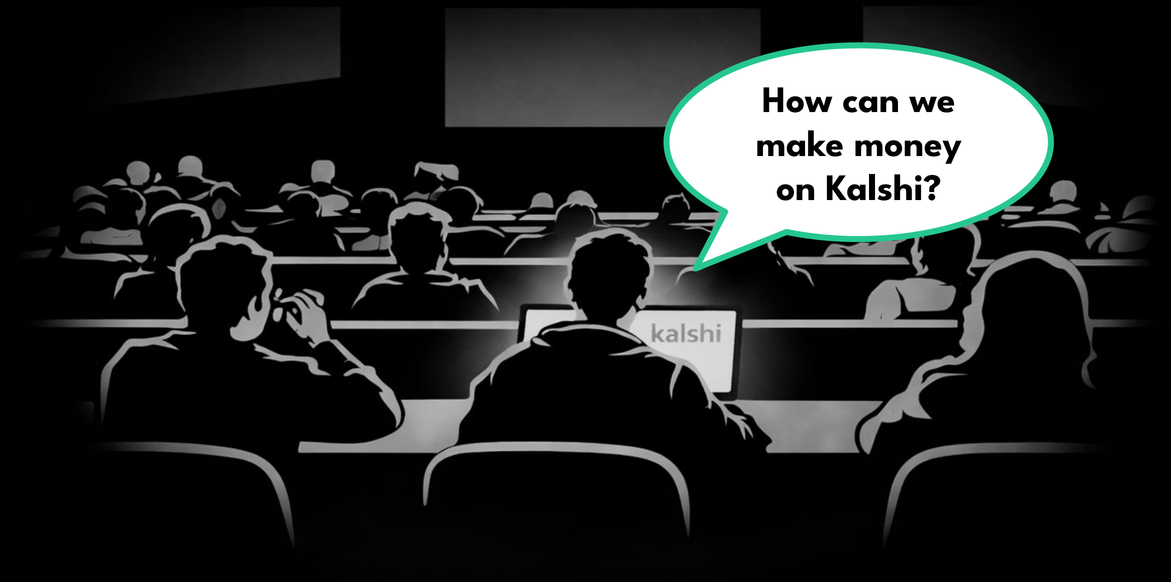
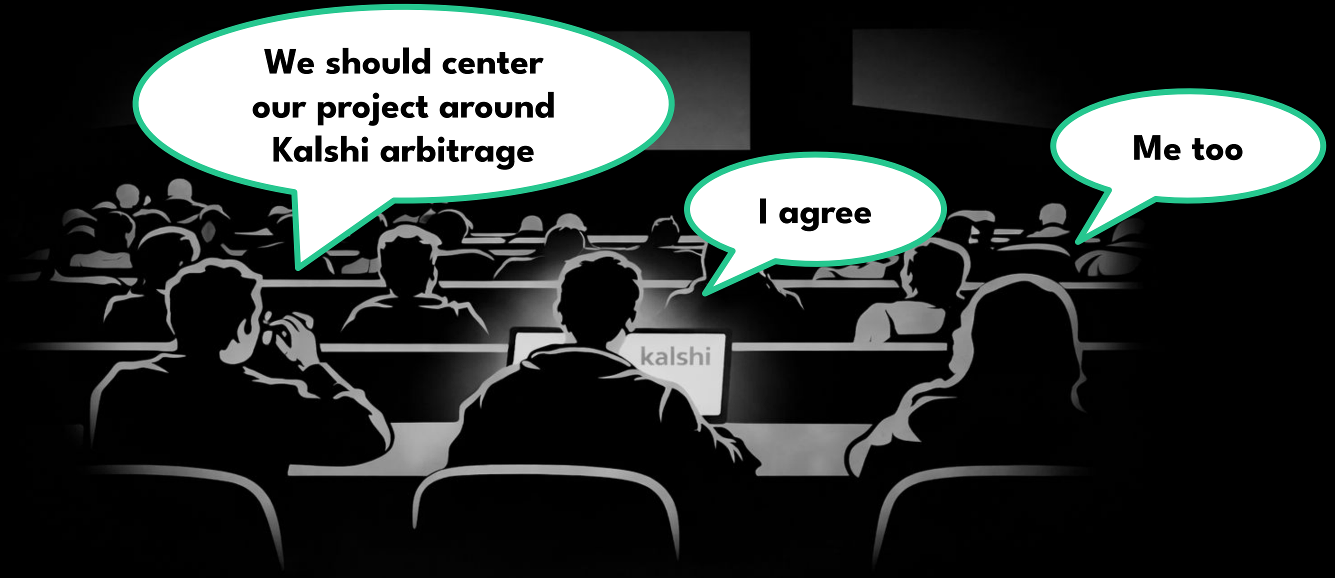
Group #1 | April 17, 2026

In our free time, we found ourselves frequently scrolling through different Kalshi markets.

As we spent time on the platform, be began to look for arbitrage opportunities on the platform.



When it came time to choose a project topic, we thought, why not explore arbitrage opportunities on Kalshi?



How Kalshi Works

"How you make money" formula

(1 - [price]) x [# contracts])

($1 - $0.66) x 1 contract

$0.34 in profit

We started noticing some markets might have strategies to guaranteed profit.

Who will win American Idol?

	Chance		
Hannah Harper	54%	Yes 54¢	No 47¢
Braden Rumfelt	13%	Yes 13¢	No 88¢
Brooks Rosser	13%	Yes 13¢	No 88¢
Jordan McCullough	13%	Yes 13¢	No 89¢

SUM: $0.93

Assumption

1 of these 4 is guaranteed to win (mutually exclusive event). Therefore, 1 of the 4 markets will payout $1 when resolving to Yes.

Cost to buy 1 Yes contract of each contestant totals to $0.93

$1.00 Payout - $0.93 Cost = $0.07 Guarantee Profit per $0.93 spent.

** This is a hypothetical example based on a current open Kalshi market.*

Project Goal

"We hope to **find and exploit market inefficiencies** to be able to profit from trading Kalshi contracts. Our final deliverable will consist of two main parts. First, we will have a report of the **strategies we tested**, the results of the strategies, and any other findings. Second, we will have a program that will **scan Kalshi markets** looking for opportunities that fit within the parameters that we found with our research and either execute orders or instantly notify us of the opportunities."

- Our Project Proposal

We tested three theses, as well as built an arbitrage scanner, with varying success

- **Strategy #1** *Exploit pricing inefficiencies in markets with fixed # of outcomes*
- **Market Scanner** *Find markets and run arbitrage math automatically*
- **Strategy #2** *Mispricing in directly related events: Crypto Range vs Threshold*
- **Strategy #3** *Exploiting inefficient 15-minute Bitcoin pricing*

STRATEGY #1

If the prices (probabilities) of a basket of markets don't sum properly, buy all of the legs.

Market	Final position	Settlement payout	Total cost	Total payout	Total return
Freestyle Skiing Men's Freeski Big Air: Medal Winner Freestyle Skiing					
Birk Ruud	3 No	$3	$0.99	$3	+$2.01 (203%)
Dylan Deschamps	3 No	$3	$2.01	$3	+$0.99 (49%)
Konnor Ralph	3 No	$3	$2.79	$3	+$0.21 (8%)
Luca Harrington	3 No	$3	$1.44	$3	+$1.56 (108%)
Mac Forehand	3 No	$0	$0.99	$0	-$0.99 (-100%)
Matej Svancer	3 No	$0	$1.44	$0	-$1.44 (-100%)
Matias Roche	3 No	$3	$2.82	$3	+$0.18 (6%)
Timothe Sivignon	3 No	$3	$2.76	$3	+$0.24 (9%)
Tormod Frostad	3 No	$0	$1.68	$0	-$1.68 (-100%)
Troy Podmilsak	3 No	$3	$1.62	$3	+$1.38 (85%)
Ulrik Samnoey	3 No	$3	$2.85	$3	+$0.15 (5%)
Total		$24	$21.39	$24	+$2.61 (12%)

The first ever example of our idea working in practice.

STRATEGY #1

We built a Kalshi market analyzer in code that had the goal of doing the profitability math including fees and slippage for us.

```
Basket | Legs | Target | Fill | MaxTop | MaxDepth |      Cost |    Payout |     Edge |     ROI
-----------------------------------------------------------------------------------------------
YES    |    9 |     10 |   10 |      2 |      248 |    $12.65 |    $10.00 |   -$2.65 | -20.95%
NO     |    8 |     10 |   10 |    107 |      291 |    $70.69 |    $70.00 |   -$0.69 |  -0.98%
Assumption: exactly one outcome resolves YES (mutually exclusive event).
Warning: NO basket uses 8/9 markets with NO asks; edge not guaranteed.
No arb (edges <= 0 at available depth).
```

```
python .\analyze_event.py KXF1RACEPODIUM-MIAGP26 --depth 4 --qty 10 --yes-winners 3
```

```
Basket | Legs | Target | Fill | MaxTop | MaxDepth |      Cost |    Payout |     Edge |     ROI
-----------------------------------------------------------------------------------------------
YES    |   22 |     10 |   10 |      5 |      419 |    $41.71 |    $30.00 |  -$11.71 | -28.07%
NO     |    7 |     10 |   10 |      5 |      393 |    $42.42 |    $40.00 |   -$2.42 |  -5.70%
Assumption: exactly 3 outcome(s) resolve YES.
Warning: NO basket uses 7/22 markets with NO asks; edge not guaranteed.
No arb (edges <= 0 at available depth).
```

STRATEGY #1

Let’s use an F1 market as an example to understand the arbitrage opportunity.

Miami Grand Prix: Winner?

Search | Top 10 Finishers | Top 5 Finishers

Andrea Kimi Antonelli	Mercedes AMG Motorsport	88%	Yes 89¢	No 19¢
George Russell	Mercedes AMG Motorsport	80%	Yes 88¢	No 20¢
Charles Leclerc	Ferrari	78%	Yes 79¢	No 28¢
Oscar Piastri	McLaren	71%	Yes 72¢	No 36¢
Lewis Hamilton	Ferrari	70%	Yes 78¢	No 30¢
Lando Norris	McLaren	63%	Yes 63¢	No 43¢
Max Verstappen	Red Bull Racing	34%	Yes 34¢	No 74¢
Pierre Gasly	Alpine F1 Team	14%	Yes 15¢	No 93¢
Isack Hadjar	Red Bull Racing	14%	Yes 15¢	No 93¢

MARKET

Will each of the following drivers be in the top 5 finishers?

STRATEGY #1

We can run our market analyzer on the F1 market example with specific parameters using flags and get immediate results.

```
python .\analyze_event.py KXF1TOP5-MIAGP26 --depth 4 --qty 1 --yes-winners 5
```

```
Basket | Legs | Target | Fill | MaxTop | MaxDepth |       Cost |     Payout |       Edge |     ROI
-----------------------------------------------------------------------------------------------------
YES    |   22 |      1 |    1 |      1 |      104 |      $6.12 |      $5.00 |     -$1.12 | -18.30%
NO     |   12 |      1 |    1 |     10 |       91 |      $7.48 |      $7.00 |     -$0.48 |  -6.42%
Assumption: exactly 5 outcome(s) resolve YES.
Warning: NO basket uses 12/22 markets with NO asks; edge not guaranteed.
No arb (edges <= 0 at available depth).
```

STRATEGY #1

We tested our strategy using real money, and saw early success.

>220	5 Yes	$0	$0.30	$0	-$0.30 (-100%)
120-139	5 Yes	$0	$0.50	$0	-$0.50 (-100%)
140-159	5 Yes	$5	$1.02	$5	+$3.98 (390%)
160-179	5 Yes	$0	$0.95	$0	-$0.95 (-100%)
180-199	5 Yes	$0	$1.06	$0	-$1.06 (-100%)
200-220	5 Yes	$0	$0.52	$0	-$0.52 (-100%)
Total		$5	$4.35	$5	+$0.65 (15%)

MARKET

How many times will Trump post on Truth Social this week.

STRATEGY #1

To expand our arbitrage opportunities, we used limit orders to avoid fees and get a better price.

We added hybrid strategies with placing resting orders between the spread

- 1. Calculated adding orders 1 tick above best bid
  - Primary motivation was to avoid Kalshi taker fees while still getting very good price
- 2. Calculated placing orders at the midpoint
  - Sacrificing price in order to improve fill likelihood
  - This generally was more successful
  - Slower and inefficient markets meant low volume so increasing fill likelihood was important

Let's look again at the F1 example, this time with our added limit order strategy.

Midpoint execution plan
Hybrid execution plan for YES basket (qty=1)

Ticker	Bid	Ask	MkPx	Spr	Queue	MakerFee	TakerFee	Save	Rec	Reason
KXF1TOP5-MIAGP26-PIA	64	72	68	8	0	$0.00	$0.02	$0.06	maker	midpoint
KXF1TOP5-MIAGP26-ALB	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-HUL	n/a	2	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-OCO	4	12	8	8	0	$0.00	$0.01	$0.05	maker	midpoint
KXF1TOP5-MIAGP26-BOR	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-ALO	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-RUS	80	88	84	8	0	$0.00	$0.01	$0.05	maker	midpoint
KXF1TOP5-MIAGP26-NOR	57	63	60	6	0	$0.00	$0.02	$0.05	maker	midpoint
KXF1TOP5-MIAGP26-COL	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-SAI	1	2	n/a	1	n/a	n/a	$0.01	n/a	taker	no midpoint
KXF1TOP5-MIAGP26-LAW	n/a	2	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-STR	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-VER	26	34	30	8	0	$0.00	$0.02	$0.06	maker	midpoint
KXF1TOP5-MIAGP26-LEC	72	79	75	7	0	$0.00	$0.02	$0.06	maker	midpoint
KXF1TOP5-MIAGP26-HAD	7	15	11	8	0	$0.00	$0.01	$0.05	maker	midpoint
KXF1TOP5-MIAGP26-BEA	2	4	3	2	0	$0.00	$0.01	$0.02	maker	midpoint
KXF1TOP5-MIAGP26-HAM	70	78	74	8	0	$0.00	$0.02	$0.06	maker	midpoint
KXF1TOP5-MIAGP26-GAS	7	15	11	8	0	$0.00	$0.01	$0.05	maker	midpoint
KXF1TOP5-MIAGP26-ANT	81	89	85	8	0	$0.00	$0.01	$0.05	maker	midpoint
KXF1TOP5-MIAGP26-PER	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-BOT	n/a	4	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid
KXF1TOP5-MIAGP26-LIN	n/a	2	n/a	n/a	n/a	n/a	$0.01	n/a	taker	no maker bid

MARKET

Will each of the following drivers be in the top 5 finishers?

STRATEGY #1

Our analyzer summarizes the arbitrage opportunities. Here's what that looks like for F1.

```
Hybrid summary for YES basket (qty=1)
Total: cost $5.28, payout $5.00, edge -$0.28, roi -5.30%
Maker: legs 11, cost $4.81, fees $0.00
Taker: legs 11, cost $0.47, fees $0.11

Hybrid summary for NO basket (qty=1)
Total: cost $6.62, payout $7.00, edge $0.38, roi 5.74%
Maker: legs 11, cost $5.62, fees $0.00
Taker: legs 1, cost $1.00, fees $0.01

Midpoint summary
Hybrid summary for YES basket (qty=1)
Total: cost $5.56, payout $5.00, edge -$0.56, roi -10.07%
Maker: legs 11, cost $5.09, fees $0.00
Taker: legs 11, cost $0.47, fees $0.11

Hybrid summary for NO basket (qty=1)
Total: cost $6.90, payout $7.00, edge $0.10, roi 1.45%
Maker: legs 11, cost $5.90, fees $0.00
Taker: legs 1, cost $1.00, fees $0.01

Recommended action now: do nothing (no positive taker edge).
```

MARKET

Will each of the following drivers be in the top 5 finishers?

While the strategy worked in theory, it didn't always work in practice. #AddARiskSystem

Trump Truth Social posts this week? (2/22-2/28)

<80	10 Yes	$0	$1.40	$0	-$1.40 (-100%)
100-119	10 Yes	$0	$2.90	$0	-$2.90 (-100%)
120-139	10 Yes	$0	$1.30	$0	-$1.30 (-100%)
140-159	10 Yes	$10	$4.00	$10.50	+$6 (133%)
160-179	10 Yes	$0	$3.00	$0	-$3 (-100%)
80-99	10 Yes	$0	$3.10	$0	-$3.10 (-100%)
Total		$10	$15.70	$10.50	-$5.70 (-36%)

STRATEGY #1

This arbitrage strategy remains profitable even this week.

Yes · <80	1¢	10	6¢	$0.60	$10 (+$9.40)	$0.10	-$0.50 (-83%)
Yes · >220	4¢	10	3.30¢	$0.33	$10 (+$9.67)	$0.40	+$0.07 (21%)
Yes · 100-119	16¢	10	18¢	$1.80	$10 (+$8.20)	$1.60	-$0.20 (-11%)
Yes · 120-139	41¢	10	18¢	$1.80	$10 (+$8.20)	$4.10	+$2.30 (128%)
Yes · 140-159	25¢	10	10.70¢	$1.07	$10 (+$8.93)	$2.50	+$1.43 (134%)
Yes · 160-179	11¢	10	11¢	$1.10	$10 (+$8.90)	$1.10	$0
Yes · 180-199	5¢	10	4¢	$0.40	$10 (+$9.60)	$0.50	+$0.10 (25%)
Yes · 200-220	4¢	10	3¢	$0.30	$10 (+$9.70)	$0.40	+$0.10 (33%)
Yes · 80-99	2¢	10	9¢	$0.90	$10 (+$9.10)	$0.20	-$0.70 (-78%)
Total		90		$8.30		$10.90	+$2.60 (33%)

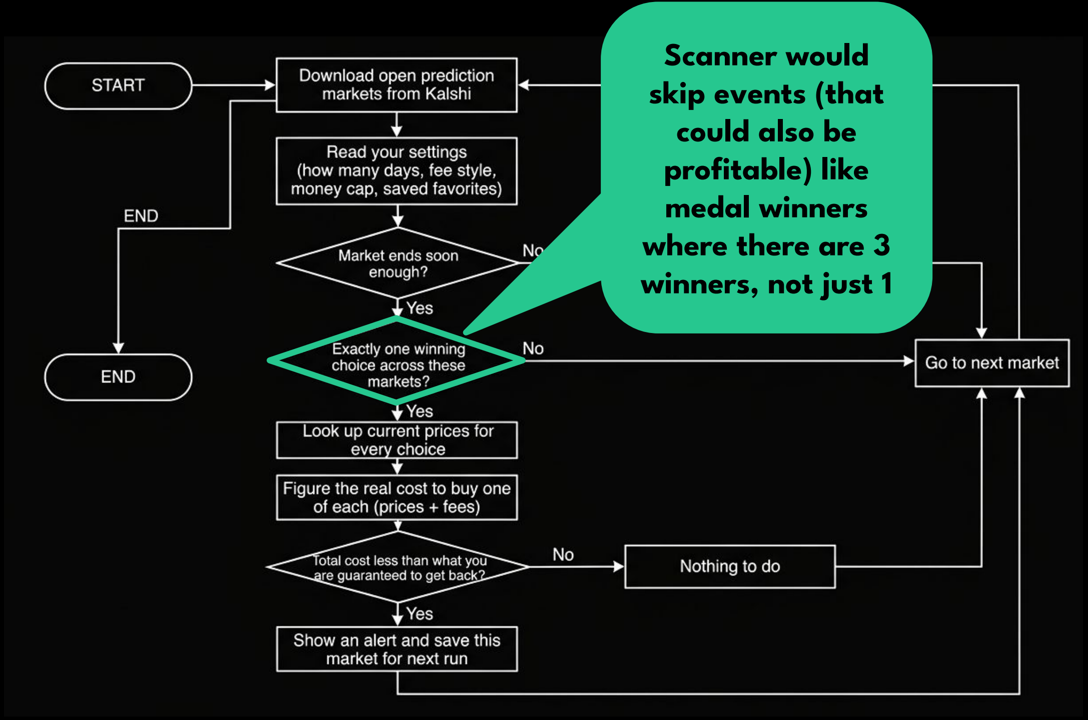
Trump Truth Social posts this week?

Currently held Kalshi Positions costing $8.30 with payout of $10 guaranteed

SCANNER

Manually searching for these rare arbitrage opportunities was tedious, so we built a scanner to find them for us.

Diagram of market scanner architecture



No limit order strategy present either

We began testing another strategy: covering all outcome possibilities by buying a combination of the two events.

Created a program that could check for possible combinations of YES/NO to buy that were poorly priced and would guarantee profit

$73,800 or above	95% ▲ 9	Yes 95¢	No 7¢
$73,900 or above	88% ▲ 12	Yes 88¢	No 13¢
$74,000 or above Current: $74,099.9188	74% ▲ 9	Yes 74¢	No 27¢
$74,100 or above	53% ▼ 1	Yes 54¢	No 47¢
$74,200 or above	35% ▲ 2	Yes 34¢	No 68¢
$74,300 or above	18% ▼ 6	Yes 18¢	No 83¢

$73,800 to 73,899.99	9% ▼ 20	Yes 8¢	No 97¢
$73,900 to 73,999.99	18% ▲ 2	Yes 15¢	No 87¢
$74,000 to 74,099.99	24% ▲ 7	Yes 26¢	No 78¢
$74,100 to 74,199.99 Current: $74,110.9002	23% ▲ 1	Yes 24¢	No 79¢
$74,200 to 74,299.99	14% ▲ 5	Yes 19¢	No 85¢
$74,300 to 74,399.99	8% ▲ 3	Yes 10¢	No 91¢

STRATEGY #2

There was no related market arbitrage, signaling that the markets were too efficient, so prices were the same.

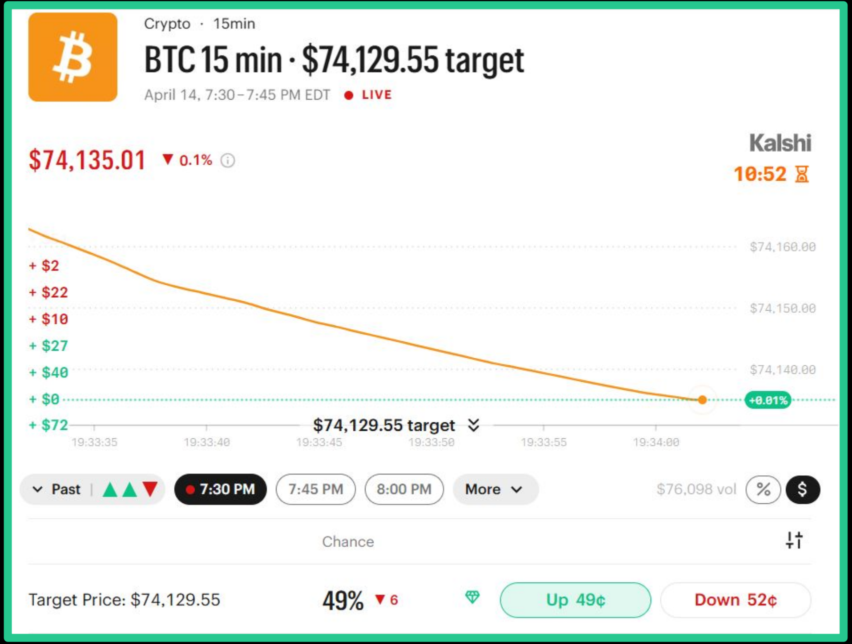
Two separate problems with this approach:

- During times of high volume (during the week), the **prices** between the two events **matched**
  - No arb opportunity
- During times of low volume (during the weekend), there was opportunity, but it was **difficult to get orders filled**
  - Created increased risk of price spikes with resting orders
  - Spreads were high

STRATEGY #3

So we tested another thesis: if we can model the price of Bitcoin using volatility, we can make money on inefficient pricing.

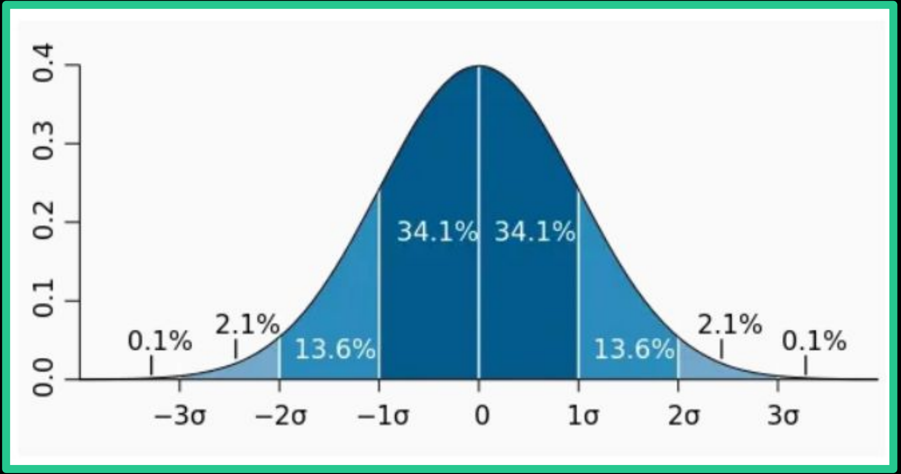
- The pricing of the contracts did not seem to always follow the price of Bitcoin
- The pricing of contracts would skew heavily in one direction with early price movement, despite Bitcoin being highly volatile
- Modeling the probability of where Bitcoin price could end up seemed easier than purely predicting price



STRATEGY #3A

First, we tried modeling Bitcoin prices with a Probability Distribution Function (PDF).

- PDF to model the price
- Mean as the current price
- Threshold as the target price
- Standard deviation based on the logarithmic returns of Bitcoin over the previous 200 minutes
    - Divided by the square root of the time remaining
    - Had to adjust the time remaining as the Kalshi Bitcoin price is based on a moving average



STRATEGY #3A

The PDF strategy was promising but still inconclusive regarding the model's profitability.

- Built a program to continuously check the Kalshi market and compare it to the model
  - Risk system does not allow entering a position if already in one
  - Can customize sizing of orders
- Through the testing we did, we found that often the supposed edge was found when either the YES or NO contract was 20-30 cents
- From a small sample size, the model is up a small amount, but more testing is needed to accurately evaluate the model
- This testing did lead to noticing a possible opportunity in price spikes

```
Time to next 15-minute event: 4.0 minutes
Current BTC price: $74941.73
Target price for next 15-minute event: $74942.58
Estimated 1-minute volatility (σ): 0.0006
Scaled volatility for 4.0 minutes (σ√T): 0.0012
Cost to buy YES at $0.72 implies an implied probability of 72.00%
Implied probability of BTC being above $74942.58 in 4.0 minutes: 49.63%
Difference between new method and implied probability from YES ask: -21.87%
✅ Order placed! ID: 485f7e7c-b178-4137-917e-6d5d66ad4cab, Status: executed
```

STRATEGY #3B

We then tested another Bitcoin strategy that exploits how Kalshi tracks moving averages.

Because retail traders tend to react slowly to large price spikes, this creates an ***opportunity*** *to enter the market early.*

- Contract prices spike and then the Kalshi graph moves
- If you can react to the data quick enough, you should be able to get in before the move
- Moves of around 10 cents are fairly common

STRATEGY #3B

Investigation into this strategy is still in progress and requires more testing and refinement.

- Created a program to subscribe to coinbase price data
- Looks for price spikes based on how close to the end of the 15 minutes it is
- Enters and then exits shortly after to secure small profits
- Needs to be deployed it to a server to minimize latency with exchange
    - Still might be too slow for a lot of movements
- However, we think this can also be used with around 1 minute left with possible market manipulation
    - This has the greatest potential if you cannot be faster than everyone else
- Currently needs more testing and refinement

```python
async def coinbase_feed():
    url = "wss://advanced-trade-ws.coinbase.com"
    sub = {
        "type": "subscribe",
        "product_ids": ["BTC-USD"],
        "channel": "market_trades",
    }
    async with aiohttp.ClientSession() as session:
        async with session.ws_connect(url) as ws:
            await ws.send_str(json.dumps(sub))
            print("Coinbase connected")
            async for msg in ws:
                if msg.type == aiohttp.WSMsgType.TEXT:
                    data = orjson.loads(msg.data)
                    for event in data.get("events", []):
                        for trade in event.get("trades", []):
                            price = float(trade["price"])
                            print(f"[coinbase] ${price:,.2f}")
                            on_price(price, time.time())
```

Next Steps

1. **Modify automatic scanner** to look for events with more than 1 yes resolution, such as “Top 5 finishing position” and take into account the limit order strategy
2. Add **active order management** and **risk system** to event arb system.
   a. Minimize adverse selection and immediate price swings
   b. Improve fill likelihood by keeping orders as best bid
   c. Immediately cross spread to secure guaranteed profit in case of price shifts or partial fills
3. **More testing** on Bitcoin pricing model to get a better sense of profitability
4. **Refine parameters** for detecting Bitcoin price spikes
   a. Deploy it to a server for minimal latency

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